

Fixed Points V: Curvature Coupling, Multi-Structure Dynamics, and Admissibility Barriers

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Part V of VI in the Fixed Point series. As both the cornerstone of the Modal Triplet Theory (MTT) collection and a stand-alone development, the series is intended to function simultaneously as a basis and as a self-contained study. Each paper in the series builds upon its predecessors, extending the fixed-point framework step by step.

Abstract

We formulate multi-structure covariance and admissibility diagnostics on the corrected FP–I–IV spine. Curvature enters through the full Laplace-type operator and its curved spectral projector, not through an unproved affine eigenvalue rule. Exact Gaussian claims are restricted to frozen linear Ornstein–Uhlenbeck systems. Their stationary covariance satisfies a Lyapunov equation, and block-diagonal damping gives a correct cross-covariance and canonical-correlation estimate. Admissibility is encoded by a finite or trace-controlled family of declared spectral, damping, and performance margins. The resulting deficit score detects exit but is neither a force nor an energy unless a separate variational source theorem is supplied. For affine Gaussian margin observables we prove finite-grid and continuous-time exit bounds. Cross-correlation controls simultaneous exits on a declared observation grid; it does not by itself prove localization, non-propagation, or selection of a post-exit state.

Revision note for this edition

Supersedes. *Fixed Points V: Curvature Coupling, Multi-Structure Dynamics, and Drivers*, version 5.

Reason. Affine curvature shifts, nonlinear Gaussian claims, an incorrectly normalized covariance bound, and a deficit score interpreted as force, energy, censorship, and post-exit selection exceeded the proved model.

Resolution. Version 6 derives covariance from the frozen linear OU system, corrects canonical-correlation bounds, declares all admissibility margins, and proves only finite-grid or continuous Gaussian exit estimates.

Retained result. Multi-structure covariance and admissibility-exit diagnostics remain rigorous on their stated domains.

Remaining boundary. Correlation does not establish propagation, localization, an exit-causing force, or selection of the next basin.

Contents

[Revision note for this edition](#)

1

1	Inherited setting and scope	2
2	Curved modal damping and exact OU modes	2
3	Multi-structure covariance	3
4	Admissibility margins and deficit score	4
5	Gaussian exit estimates	5
6	Simultaneous exits and the limit of correlation data	5
7	Conclusion	6

1 Inherited setting and scope

Let $I = [t_0, t_1]$ be a finite parameter interval and let (B, g_t) be a Riemannian base with uniformly bounded geometry on I . This formulation does not identify the FP stabilization parameter with physical Lorentzian time. Existence of a projected fixed point and promotion to a full equilibrium are inherited only when the hypotheses of FP I and FP II hold. Disturbance floors are interpreted according to FP III, and curvature uses the full operator and projector construction of FP IV.

At a frozen configuration x , write

$$L_x = \nabla_x^* \nabla_x + \mathcal{R}_x, \quad P_x = P_{\mathcal{R}}(x), \quad Q_x = I - P_x.$$

The contour defining P_x is assumed to remain in the resolvent set. If the old uncurved projector is used instead, the leakage term $Q\mathcal{R}P$ from FP IV must be retained.

All matrix theorems below are finite dimensional. They also extend to a separable Hilbert space when the noise covariance is trace class, the semigroup is exponentially stable, and every displayed covariance product is well defined. Modal sums are never used without a finite truncation or an explicit summability hypothesis.

2 Curved modal damping and exact OU modes

Let α denote a joint spectral label, including multiplicity. In a frozen Q_x -sector linearization, suppose the scalar mode satisfies the Itô equation

$$da_\alpha = -\gamma_\alpha(x)a_\alpha dt + \sqrt{q_\alpha(x)} dW_\alpha(t), \quad \gamma_\alpha(x) > 0. \quad (2.1)$$

Here γ_α is the damping margin obtained after the one-sided nonlinear estimate; it is not a shared deterministic/stochastic disturbance parameter. Curvature dependence is inherited from L_x . An affine response formula for an eigenvalue is an additional approximation, not a consequence of the Weitzenböck identity.

Lemma 2.1 (Exact scalar OU variance). *Equation (2.1) has the unique invariant law*

$$\mathcal{N}\left(0, \frac{q_\alpha}{2\gamma_\alpha}\right).$$

Deterministic forcing f_α instead gives the input-to-state scale $|f_\alpha|/\gamma_\alpha$; these two quantities are not interchangeable.

Proof. The variation-of-constants formula gives

$$a_\alpha(t) = e^{-\gamma_\alpha t} a_\alpha(0) + \sqrt{q_\alpha} \int_0^t e^{-\gamma_\alpha(t-s)} dW_\alpha(s).$$

Itô isometry yields variance $q_\alpha(1 - e^{-2\gamma_\alpha t})/(2\gamma_\alpha)$, which converges to the displayed value. The deterministic scale follows by integrating the stable kernel against a bounded input. $\square$

3 Multi-structure covariance

Collect a finite set of frozen linearized amplitudes in $a(t) \in \mathbb{R}^d$:

$$da = -\Gamma a dt + B dW_t, \quad Q := BB^\top, \quad (3.1)$$

where $\Gamma = \Gamma^\top \succeq \gamma_0 I$ with $\gamma_0 > 0$.

Theorem 3.1 (Stationary covariance). *Equation (3.1) has a unique centered Gaussian invariant law with covariance*

$$\Sigma = \int_0^\infty e^{-t\Gamma} Q e^{-t\Gamma} dt.$$

It is the unique solution of

$$\Gamma \Sigma + \Sigma \Gamma = Q, \quad \|\Sigma\|_{\text{op}} \leq \frac{\|Q\|_{\text{op}}}{2\gamma_0}.$$

Proof. Exponential stability makes the integral converge. Differentiating the integrand proves the Lyapunov equation. Uniqueness follows by applying the stable semigroup to the homogeneous equation, and the norm estimate follows from $\|e^{-t\Gamma}\| \leq e^{-\gamma_0 t}$. $\square$

Partition the structures into A and B and assume for this section that

$$\Gamma = \text{diag}(\Gamma_A, \Gamma_B), \quad \Gamma_A \succeq \gamma_A I, \quad \Gamma_B \succeq \gamma_B I.$$

Then the cross block obeys the Sylvester equation

$$\Gamma_A \Sigma_{AB} + \Sigma_{AB} \Gamma_B = Q_{AB}.$$

Lemma 3.2 (Cross-covariance bound). *Under the block-damping hypothesis,*

$$\|\Sigma_{AB}\|_{\text{op}} \leq \frac{\|Q_{AB}\|_{\text{op}}}{\gamma_A + \gamma_B}.$$

In particular, $Q_{AB} = 0$ implies $\Sigma_{AB} = 0$ in this frozen linear model.

Proof. The Sylvester solution is $\Sigma_{AB} = \int_0^\infty e^{-t\Gamma_A} Q_{AB} e^{-t\Gamma_B} dt$. Taking norms proves the claim. $\square$

Definition 3.3 (Canonical cross-correlation). For a positive semidefinite covariance block matrix, define

$$\rho_{AB} := \|\Sigma_A^{\dagger/2} \Sigma_{AB} \Sigma_B^{\dagger/2}\|_{\text{op}}.$$

The covariance support inclusions imply $0 \leq \rho_{AB} \leq 1$. Let $m_A, m_B > 0$ denote the smallest positive eigenvalues of Σ_A, Σ_B on their supports.

Theorem 3.4 (Damping bound for canonical correlation). *Under the preceding hypotheses,*

$$\rho_{AB} \leq \frac{\|Q_{AB}\|_{\text{op}}}{(\gamma_A + \gamma_B)\sqrt{m_A m_B}}.$$

Consequently the right-hand side being smaller than $\rho_0 \in (0, 1)$ is a sufficient condition for $\rho_{AB} < \rho_0$.

Proof. Use submultiplicativity, the identities $\|\Sigma_A^{\dagger/2}\| = m_A^{-1/2}$ and $\|\Sigma_B^{\dagger/2}\| = m_B^{-1/2}$, and Lemma 3.2. $\square$

The smallest positive covariance eigenvalues are essential. Replacing them by $\|\Sigma_A\|$ and $\|\Sigma_B\|$ reverses the relevant normalization and does not bound canonical correlation.

4 Admissibility margins and deficit score

Fix a finite family of continuous margins $m_j(x)$, $j \in \mathcal{J}$. They may include:

- separation of the curved low spectral cluster from the rest of the spectrum;
- positive one-sided damping margins $\gamma_\alpha(x)$;
- declared deterministic performance bounds $r_\alpha^{\text{det}} - |f_\alpha|/\gamma_\alpha$;
- declared stochastic variance bounds $r_\alpha^{\text{stoch}} - q_\alpha/(2\gamma_\alpha)$.

For an infinite family, replace finiteness by uniform convergence and compact control sufficient to make the infimum below continuous.

Definition 4.1 (Admissible domains and deficit score). Set

$$\mathcal{D}_\varepsilon := \{x : m_j(x) \geq \varepsilon \text{ for every } j \in \mathcal{J}\}, \quad \mathfrak{B}(x) := \max_{j \in \mathcal{J}} [-m_j(x)].$$

Then $x \in \mathcal{D}_0$ exactly when $\mathfrak{B}(x) \leq 0$, and $x \in \mathcal{D}_\varepsilon$ exactly when $\mathfrak{B}(x) \leq -\varepsilon$.

Proposition 4.2 (Diagnostic status). *The score $\mathfrak{B}$ is an exact scalar encoding of the declared margin constraints. It is not inserted into the evolution equation. It is not a physical energy, selection potential, or infinite enforcement cost unless an independent variational source theorem identifies it with such an object.*

Theorem 4.3 (Deterministic exit detection). *Let $X : I \rightarrow B$ be continuous and suppose the margins are continuous. If $X(t_0) \in \mathcal{D}_0$ and $\mathfrak{B}(X(t_1)) > 0$, there is a first exit time*

$$\tau := \inf\{t \in [t_0, t_1] : \mathfrak{B}(X(t)) > 0\},$$

with $\mathfrak{B}(X(\tau)) = 0$. At times with $\mathfrak{B}(X(t)) > 0$, at least one declared constraint fails.

Proof. The maximum of finitely many continuous functions is continuous. The result follows from the intermediate value theorem and the definition of first exit. $\square$

This theorem detects loss of the declared description. It neither supplies a force causing the exit nor selects the state or basin after exit.

5 Gaussian exit estimates

We now impose the stronger structure actually needed for Gaussian claims. Assume the stationary solution of (3.1) and an exact affine linearized margin model

$$m_j(t) = \bar{m}_j + \ell_j^\top a(t), \quad \bar{m}_j > 0. \quad (5.1)$$

Thus $Z_j(t) := \ell_j^\top a(t)$ is a centered stationary Gaussian process with

$$v_j := \text{Var } Z_j(t) = \ell_j^\top \Sigma \ell_j \leq \|\ell_j\|^2 \frac{\|Q\|}{2\gamma_0}.$$

A merely Lipschitz nonlinear function of a Gaussian process is not generally Gaussian, so (5.1) cannot be replaced by Lipschitz regularity alone.

Theorem 5.1 (Finite-grid exit bound). *For observation times $t_1, \dots, t_N$,*

$$\mathbb{P}\left(\min_{j \in \mathcal{J}, 1 \leq k \leq N} m_j(t_k) < 0\right) \leq \sum_{j \in \mathcal{J}} N \exp\left(-\frac{\bar{m}_j^2}{2v_j}\right),$$

with a zero-variance term interpreted as zero when $\bar{m}_j > 0$.

Proof. For each pair (j, k) , the Gaussian tail bound gives $\mathbb{P}(Z_j(t_k) < -\bar{m}_j) \leq \exp[-\bar{m}_j^2/(2v_j)]$. Apply the union bound. No temporal independence is required. $\square$

Theorem 5.2 (Continuous-time Gaussian exit bound). *Assume each Z_j is separable with almost surely continuous paths and*

$$e_j := \mathbb{E} \sup_{t \in I} [-Z_j(t)] < \infty, \quad \sigma_j^2 := \sup_{t \in I} \text{Var } Z_j(t).$$

If $\bar{m}_j > e_j$ for every j , then

$$\mathbb{P}\left(\inf_{t \in I} \min_{j \in \mathcal{J}} m_j(t) < 0\right) \leq \sum_{j \in \mathcal{J}} \exp\left[-\frac{(\bar{m}_j - e_j)^2}{2\sigma_j^2}\right].$$

Proof. Apply the Borell–TIS inequality to the separable Gaussian process $-Z_j$ and then take a union bound over the finite constraint family. $\square$

Finiteness of e_j can be verified from the canonical increment metric by a metric-entropy bound. Exponential covariance decay alone suggests the usual $\sqrt{\log(1 + \gamma_0|I|)}$ scale, but that scale is not asserted without the required entropy estimate.

6 Simultaneous exits and the limit of correlation data

For two scalar affine margin observables at a fixed time, standardize the exit variables as $U = -Z_A/\sigma_A$ and $V = -Z_B/\sigma_B$. Suppose $-1 < \rho_0 < 1$, $\text{Corr}(U, V) \leq \rho_0$, and the standardized positive margins obey $u_A = \bar{m}_A/\sigma_A$, $u_B = \bar{m}_B/\sigma_B$.

Lemma 6.1 (Fixed-time simultaneous-exit bound).

$$\mathbb{P}(U \geq u_A, V \geq u_B) \leq \exp\left[-\frac{(u_A + u_B)^2}{4(1 + \rho_0)}\right].$$

For N declared observation times at which the same bounds hold, the probability of a simultaneous observed exit is at most N times the right-hand side.

Proof. The joint event implies $U + V \geq u_A + u_B$, while $\text{Var}(U + V) \leq 2(1 + \rho_0)$. Apply the one-dimensional Gaussian tail bound and then the union bound over observation times. $\square$

Proposition 6.2 (No propagation theorem from covariance alone). *Equal-time covariance or canonical correlation does not determine causal propagation, temporal clustering, or a post-exit transition. Such conclusions require a specified coupled evolution or transition kernel and control of cross-time covariance. Therefore FP V makes no probabilistic censorship or non-propagation claim from ρ_{AB} alone.*

7 Conclusion

FP V now supplies a rigorous frozen-linear covariance layer and a correctly scoped admissibility layer. Curvature is handled through the FP IV full operator and projector. Joint modes use the FP II/III indexing and disturbance conventions. Canonical correlation is normalized by the smallest positive covariance eigenvalues, and Gaussian persistence is proved only for affine Gaussian margin observables under the assumptions required by the relevant concentration theorem. The deficit score detects a declared loss of admissibility but is not promoted to dynamics, energy, propagation, or post-exit selection.

References

- [1] G. Da Prato and J. Zabczyk, *Stochastic Equations in Infinite Dimensions*, Encyclopedia of Mathematics and Its Applications, Vol. 44, Cambridge University Press, Cambridge, 1992.
- [2] R. Temam, *Infinite-Dimensional Dynamical Systems in Mechanics and Physics*, Applied Mathematical Sciences, Vol. 68, Springer-Verlag, New York, 1997.